



MFU ACCESS POINTS ENERGY SAVING

THICHAKORN WITCHAYAKUL

FANA YAHLEE

ROMTHEERA WANG

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2023

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

MFU ACCESS POINTS ENERGY SAVING

THICHAKORN WITCHAYAKUL

FANA YAHLEE

ROMTHEERA WANG

A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2023

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

MFU ACCESS POINTS ENERGY SAVING

THICHAKORN WITCHAYAKUL

FANA YAHLEE

ROMTHEERA WANG

THIS SENIOR PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

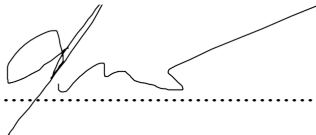
2023

EXAMINING COMMITTEE



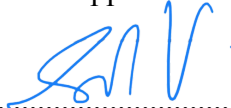
ADVISOR

(Aj. Dr.Mahamah Sebakor)



COMMITTEE

(Asst. Prof. Dr.Suppakarn Chansareewittaya)



COMMITTEE

(Asst. Prof. Dr.Surapol Vorapatratorn)

ACKNOWLEDGEMENT

This project will not be successful without the kind support of the advisor, committees and all supportive persons. We would like to thank everyone for their advice and help. We are also grateful to our parents and friends for their continuous encouragement throughout this project. This accomplishment would not have been possible without them. Thank you.

Thichakorn Witchayakul

Fana Yahlee

Romtheera Wang

Title	MFU ACCESS POINTS ENERGY SAVING	
Author	Miss Thichakorn	Witchayakul
	Miss Fana	Yahlee
	Miss Romtheera	Wang
Degree	Bachelor of Engineering (Computer Engineering)	
Supervisory Committee	Aj. Dr.Mahamah Sebakor	Advisor
	Asst. Prof. Dr.Suppakarn Chansareewittaya	Committee
	Asst. Prof. Dr.Surapol Vorapatratorn	Committee

Abstract

Access point wireless is a system that controls the use of access points in Mae Fah Luang University. It detects the amount of devices that gain access to the access points. If the detected number appears to be zero, the system will automatically turn the access points off to save electricity and costs in the university. There's also an alternative way of turning the access points on and off manually in case that the automatic system is not comprehensive enough in some scenarios.

Keyword: Access points, wireless network

CONTENTS

ACKNOWLEDGEMENT	3
Abstract	4
LIST OF TABLES	6
LIST OF FIGURES	7
CHAPTER 1 INTRODUCTION	8
1.1 Background and Rationale	8
1.2 Objective	8
1.3 Scope	8
1.4 Methodology	8
1.5 Plan	9
1.6 Expected Result	9
1.7 Place and Equipment	9
CHAPTER 2 LITERATURE REVIEW	10
2.1 Related Theory	10
2.2 Related Technology	10
CHAPTER 3 ANALYSIS AND DESIGN	13
3.1 Block diagram	13
3.2 Website Design	14
CHAPTER 4 EXPERIMENT AND RESULTS	16
4.1 Track and display graph on the website	17
4.2 Experiment	17
4.3 Experiment Result	17
CHAPTER 5 CONCLUSION	18
4.1 Conclusion	18
4.2 Discussion	18
REFERENCES	19
APPENDICES	20

LIST OF TABLES

Table	Page
Table 1.1 Working plan	9
Table 1.2 Result of Experiment	17

LIST OF FIGURES

Figures	Page
Figure 2.2.3 Access point	10
Figure 2.2.4 VMware Workstation	11
Figure 2.2.5 Visual Studio	12
Figure 2.2.6 Ubuntu	12
Figure 3.2.1 Login Page	14
Figure 3.2.2 AP Page	14
Figure 3.2.3 Savings Page	15
Figure 3.2.4 Savings Page	15
Figure 4.1.2 Power Savings	16

CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Today, access point antennas in Mae Fah Luang University are being used extravagantly due to the approach of turning them on and off manually. Many times the density of devices asking for connection from access points is rather small in particular times and areas, often, there's no request to connect at all, which makes turning on access points in these scenarios become a waste for both energy and money.

To prevent the problem from continuing any longer, MFU ACCESS POINTS ENERGY SAVING is one of the efficient problem-solving methods. The brief function of the system is to automatically turn on and off the access points under the condition that the bandwidth must be enough. The 5GHz network will be turned off once devices in the area are detected to be zero or the small number of devices in the area are able to access other access points nearby using 2.4GHz network as a detector. This way access points antennas in Mae Fah Luang University are being used as its best.

1.2 Objective

1.2.1 To obtain the system which reducing electric consumption in wireless network

1.3 Scope

1.3.1 General user

1.3.1.1 Mae Fah Luang University's staff

1.3.1.2 Mae Fah Luang University's students

1.3.2 Area of work

1.3.2.1 Mae Fah Luang University

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 System analysis

1.4.4 System design

1.4.5 Senior project 1 document preparation

1.4.6 Senior Project 1 exam

1.4.7 System development

1.4.8 System testing

1.4.9 Senior Project 2 document preparation

1.4.10 Senior Project 2 exam

1.5 Plan

Table 1.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Senior project 1 document preparation										
Senior Project 1 exam										
System development										
System testing										
Senior project 2 document preparation										
Senior Project 2 exam										

1.6 Expected Result

- 1.6.1 The system can run perfectly fine
- 1.6.1 The system reduces the energy wasted

1.7 Place and Equipment

1.7.1 place

1.7.1.1 Mae Fah Luang University

1.7.2 Equipment

1.7.2.1 Hardware

- Access point
- Router
- Laptop

1.7.2.2 Software

- VMware Workstation 15.5.2
- Visual Studio version 1.76.2
- Ubuntu

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

The energy-saving system under development operates by automatically controlling the activation and deactivation of the 5.0G network based on the occupancy status of each access point which uses the 2.4G network as a detector. When the access point is unoccupied or the occupation's numbers are low and can be connected to the nearby access points instead, the system initiates automatically shutdown the 5.0G network of this access point to save energy. Conversely, when someone is connected to the access point, the system activates the 5.0G network and will keep activated if a decent amount of people are connected and can't find nearby access points to connect to.

2.2 Related Technology

2.2.1 Laptop

2.2.2 Router is a networking device that serves as a central point of communication between multiple devices or networks.

2.2.3 Access point is a networking device that allows wireless devices to connect to a wired network.



figure 2.2.3 Access point

2.2.4 VMware Workstation is a popular virtualization software application developed by VMware Inc. It allows users to create and run multiple virtual machines (VMs) on a single physical computer. With VMware Workstation, you can install and run different operating systems simultaneously.

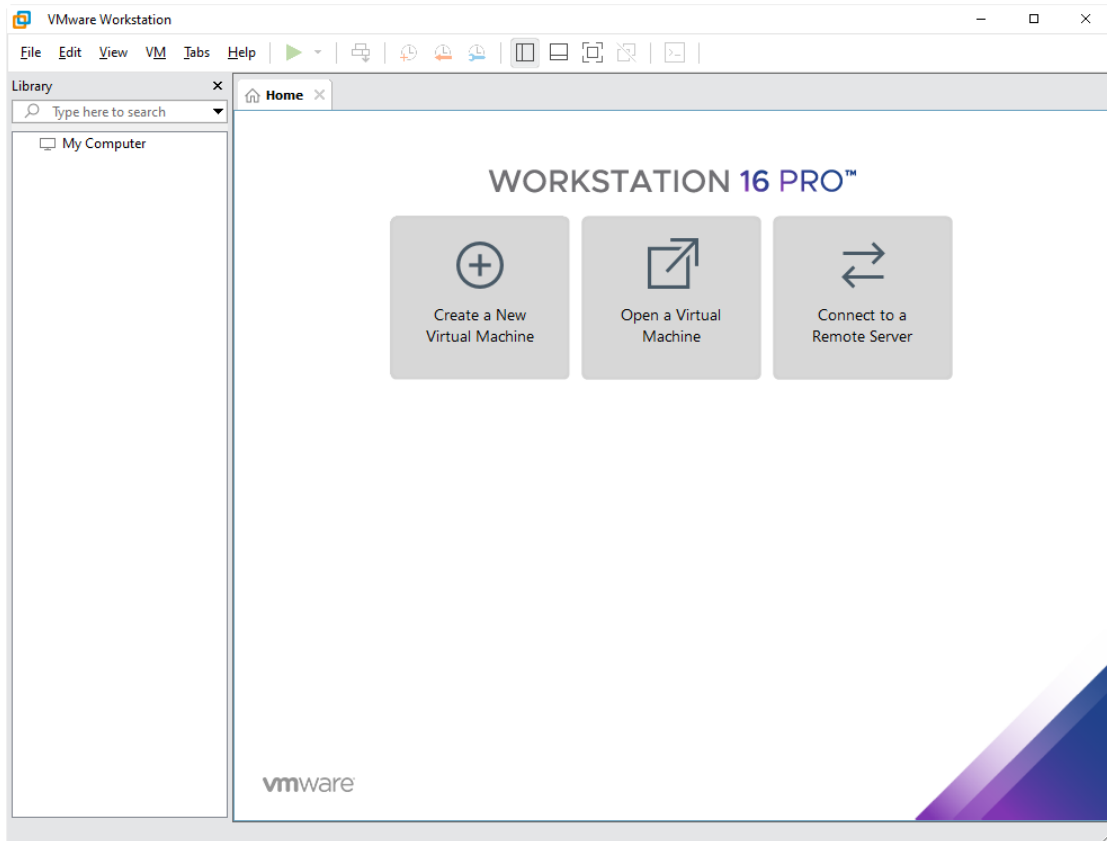


figure 2.2.4 VMware Workstation

2.2.5 Visual Studio is an integrated development environment (IDE) created by Microsoft. It is primarily used for developing computer programs, websites, web applications, and mobile applications. Visual Studio provides a comprehensive set of tools and features that assist developers in writing, debugging, and deploying software solutions.



figure 2.2.5 Visual Studio

2.2.6 Ubuntu is the modern, open source operating system on Linux for the enterprise server, desktop, cloud, and IoT.

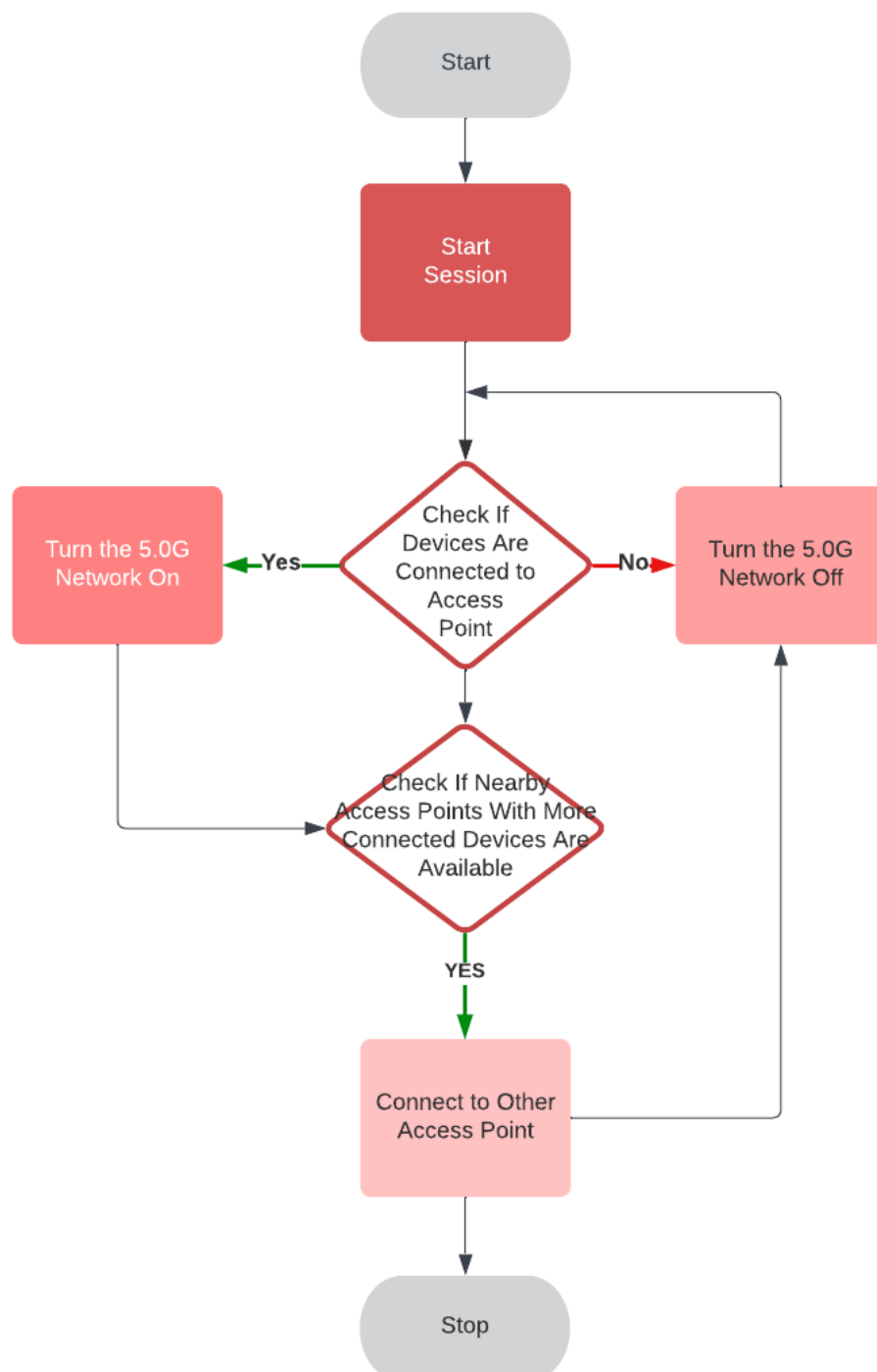


figure 2.2.6 Ubuntu

CHAPTER 3 ANALYSIS AND DESIGN

3.1 Block Diagram

The energy-saving system is visually represented by a formal Block Diagram, which illustrates the relationships and interconnections among its components. This diagram provides a clear and concise overview of the system's structure and functioning.



3.2 Website Design

3.2.1 Login Page

This webpage is designed for the users to login.



figure 3.2.1 Login Page

3.2.2 AP Page

This webpage is designed for the users to keep track of 2.4GHz and 5.0GHz status.

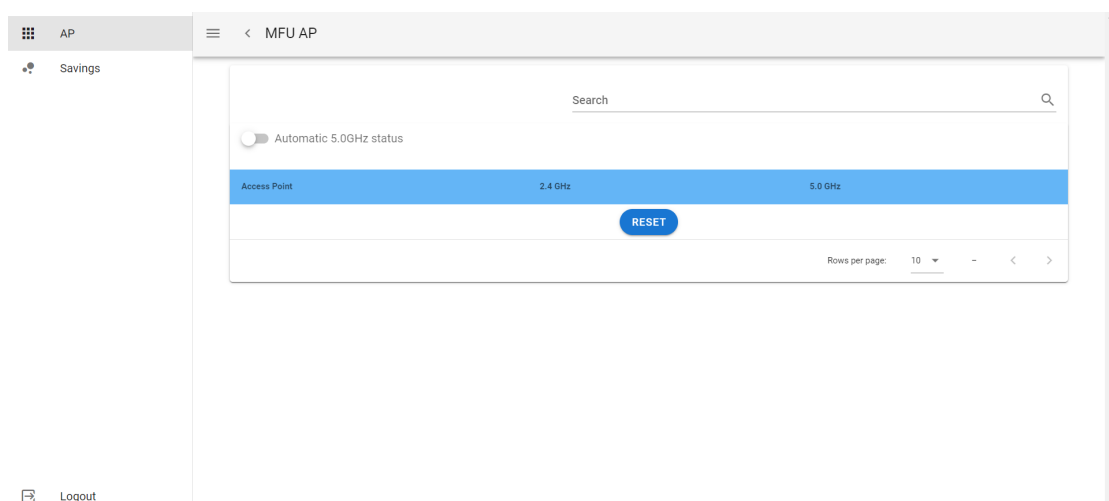


figure 3.2.2 AP Page

3.2.3 Savings Page

This webpage is designed for the users to keep track of energy and money savings.

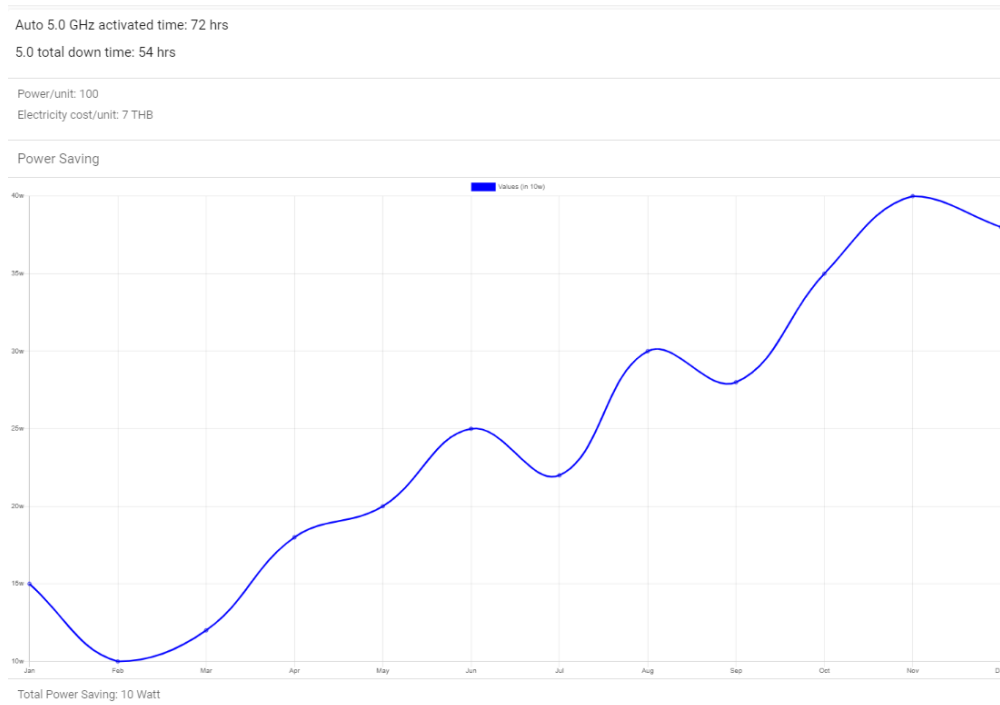


figure 3.2.3 Savings Page

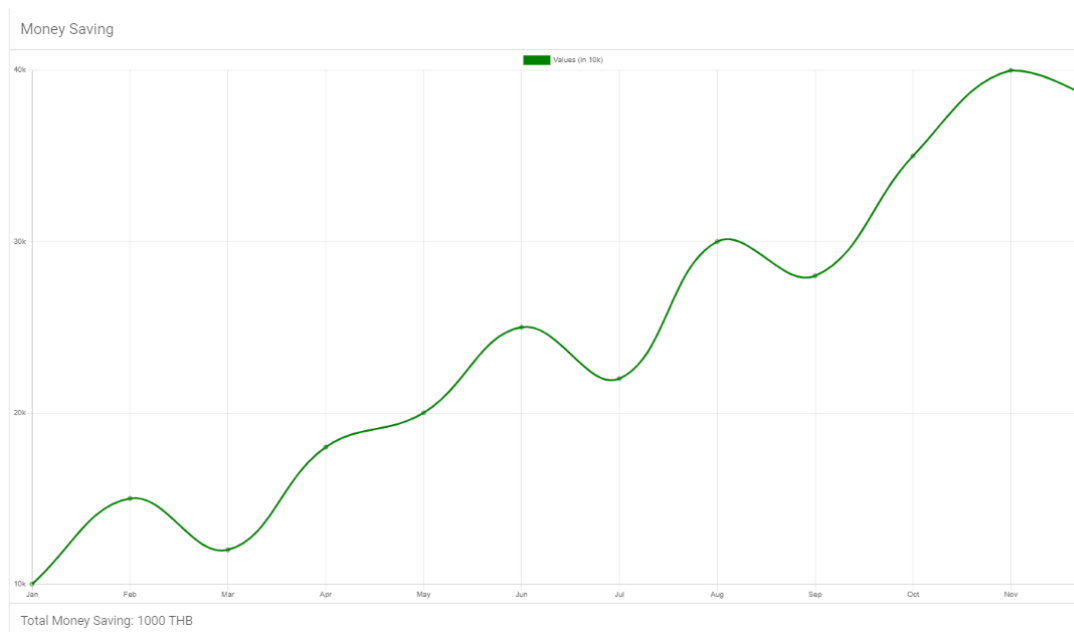


figure 3.2.4 Savings Page

CHAPTER 4

EXPERIMENT AND RESULTS

4.1 Track and display graph on the website

After logging into the website, on the savings page, there are two graphs used to track the amount and energy savings and money savings.

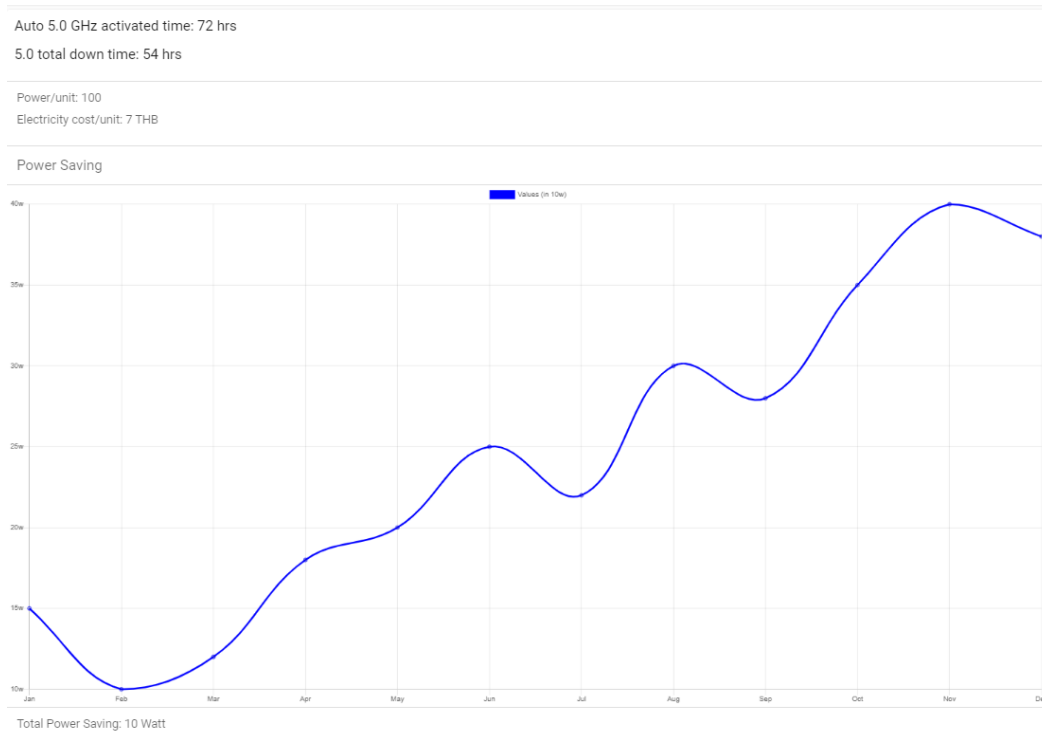


figure 4.1.2 Power Savings

4.2 Experiment

4.2.1 Tested by turning on the 'Automatic 5.0GHz Status' on the AP page to activate the code for turning on and off 5.0GHz based on given conditions.

4.2.2 Continue to the Savings page and observe the results

4.3 Experiment Result

4.3.1 Wait for three days when 'Automatic 5.0GHz Status' is on.

Day	Energy Savings	Money Savings
1		
2		
3		

Table 1.1 Result of Experiment

After testing the system, the energy and money that could have been saved if the 5.0GHz is only active based on conditions are revealed.

CHAPTER 5

CONCLUSION

5.1 Conclusion

In conclusion, implementing energy-saving measures in MFU Access Points offers multiple benefits, including improved energy efficiency, reduced environmental impact, cost savings, and optimized network performance. These measures not only benefit organizations and businesses but also contribute to a more sustainable and eco-friendly future.

5.2 Discussion

This project is providing cost savings, environmental sustainability, and optimized network performance. These measures are made possible by technological advancements, collaboration among stakeholders, and the potential for future innovations.

REFERENCES

- [1] “Cisco Learning Network” [Online]. Available:
<https://learningnetwork.cisco.com/s/>. [Accessed: 22-July-2023]
- [2] “Cisco Wireless Access Point ” [Online]. Available:
<https://www.cisco.com/c/en/us/solutions/small-business/resource-center/networking/what-is-access-point.html>. [Accessed: 25-July-2023]
- [3] “Wireless Access Point ” [Online]. Available:
<https://www.techtarget.com/searchmobilecomputing/definition/access-point>. [Accessed: 25-July-2023]
- [4] “Netmiko” [Online]. Available:
<https://pypi.org/project/netmiko/>. [Accessed: 01-Oct-2023]
- [5] “Visual studio Code” [Online]. Available:
<https://code.visualstudio.com/learn>. [Accessed: 01-Nov-2023]
- [6] “Vue.js” [Online]. Available:
<https://vuejs.org/guide/quick-start.html>. [Accessed: 02-Nov-2023]
- [7] “GitHub” [Online]. Available:
<https://github.com/>. [Accessed: 02-Nov-2023]
- [8] “MySQL” [Online]. Available:
<https://www.mysql.com/>. [Accessed: 02-Nov-2023]

APPENDICES

```
#Shut down, bring up 5.0GHz code

import netmiko
import time
import getpass
from netmiko import ConnectHandler

# Create a dictionary with the device details
device = {
    'device_type': 'cisco_ios',
    'host': '172.30.99.56',
    'username': 'Admin',
    'password': 'CITS@WLC2023',
    'secret': 'CITS@WLC2023',
    'fast_cli': True,
}

# Create a list of configuration commands for shutting down the interface
shutdown_commands = [
    'configure terminal',
    'ap dot11 5ghz shutdown',
    'exit'
]

# Create a list of configuration commands for bringing up the interface
no_shutdown_commands = [
    'configure terminal',
    'ap dot11 5ghz no shutdown',
    'exit',
]

# SSH Show version/uptime
try:
    with netmiko.ConnectHandler(**device) as net_connect:
        # Send command to show version and uptime
        output = net_connect.send_command('show version | include uptime')
        print(output.strip())

        # Enable privilege mode
        net_connect.enable()

        #Send command to show running configuration
        result = net_connect.send_command('show interface summary')
        print(result)

        # Enter configure mode
```

```
net_connect.config_mode()

# Send configuration commands
output = net_connect.send_config_set(config_commands)
print(output)

# Exit configure mode
net_connect.exit_config_mode()

# Save the configuration
output = net_connect.send_command_expect('write memory')
print(output)

# Close the SSH connection
net_connect.disconnect()

except netmiko.ssh_exception.NetMikoAuthenticationException:
    print("Authentication failed. Please check your credentials.")
except netmiko.ssh_exception.NetMikoTimeoutException:
    print("Device connection timed out. Please ensure the device is reachable.")
except Exception as e:
    print("An error occurred:", str(e))
```